



Bethina Joshitha Chowdhary
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Female
DOB: 11/08/2003

Examination	University	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	Narayana High School	2021	93.40%
Matriculation	CBSE	Narayana High School	2019	92.80%

SCHOLASTIC ACHIEVEMENTS

- Achieved **All India Rank 148** among 141K+ aspirants in **JEE ADVANCED 2021** (2021)
- Achieved **All India Rank 365** among 1.05M+ aspirants in **JEE MAINS 2021** (2021)
- Achieved **100 percentile** in **Physics** and an overall **99.97 percentile** in **JEE MAINS** (2021)
- Recipient of the prestigious **KVPY Fellowship** by **IISc Bangalore** with an **All India Rank** of **275** (2020)
- Secured **Rank 43** in **AP-EAPCET** out of **1.66 lakh** candidates conducted by **APSCE** (2021)
- Secured **Rank 187** in **TS-EAMCET** out of **2.27 lakh** candidates conducted by **TSCHE** (2021)

PROFESSIONAL EXPERIENCE

Software Engineer | AlloyDB, Google Cloud

(Summer 2024)

Summer Internship

- Developed an **Anomaly Detection** model for some metrics of the monitoring system of **AlloyDB** by creating testing sets and integrated it with Google Visualization Studio to view the existing anomalies on the **Dashboard**
- Utilized a structural **Bayesian time-series** approach to enhance anomaly detection accuracy and system efficiency
- Enabled faster resolution of **customer issues** for AlloyDB on-callers by pinpointing the abnormal metrics

KEY PROJECTS

Movie Recommendation System | Course Project

(Autumn 2023)

Guide: Prof. Preethi Jyothi, Course: Artificial Intelligence and Machine Learning

- Recommends new movies for a user based on their viewing habits and ratings or top movies similar to a given one
- Used **content-based** similarity, which suggests films by calculating the cosine-similarity of the genre composition
- Created a hybrid model using **SVD**, **KNN** and obtained a final **NDCG** metric measure of **0.97** for the test data

Simulation of a P2P Cryptocurrency Network | Course Project

(Spring 2024)

Guide: Prof. Vinay Ribeiro, Course: Introduction to Blockchains, Cryptocurrencies and Smart Contracts

- Built a discrete-event simulator for a **P2P** network wherein each peer generates **transactions** randomly in time
- Included Loop-less transaction forwarding, POW, transaction verification, and blockchain forks, as in **bitcoin**
- Incorporated node hyperparameters to analyze network during a **Double Selfish Mining Attack** by two miners

Portal for the courses | Course Project

(Autumn 2022)

Guide: Prof. Kavi Arya, Course: Software Systems Lab

- Developed an online **Learning Management System(LMS)** to host online courses using **Django** framework
- Created **Role based Authentication** with Login/Sign-Up facility using **SQLite Database** in backend
- Implemented functionalities for instructors to create **courses**, upload **assignments**, manage student **grades**

Compiler Construction | Course Project

(Spring 2024)

Guide: Prof. Uday Khedker, Course: Implementation of Programming Languages Lab

- Built a **compiler** for C-like Language called **SCLP** by constructing **scanner**, **parser** using flex, yacc and C++
- Incorporated **semantic checks**, constructed **symbol tables**, and obtained the Abstract Syntax Tree (**AST**)
- Extended it to support functions, **register allocation** and the generation of **RTL** and **MIPS assembly code**

Memory/Cache hierarchy optimizations for SAT solvers | Course Project

(Spring 2023)

Guide: Prof. Biswabandan Panda, Course: Digital Logic Design and Computer Architecture

- Successfully implemented Inclusive, Exclusive and **NINE cache hierarchy policies** in **champsim** simulator and found that **L1-L2 Exclusive** and **L2-LLC Exclusive** gives highest number of Instructions per cycle(**IPC**)
- Optimized the performance by evaluating with various **replacement policies** like FIFO, RAND, LFU, LRU
- Found that **RAND** is the most optimal replacement policy for SAT solvers with IPC increase of **0.81%** from LRU

Image Processing | Course Project

(Autumn 2022)

Guide: Prof. Suyash.P.Awate, Course: Data Analysis and Interpretation

- Implemented **Principle Component Analysis (PCA)** for **Hyperplane Fitting** and **Linear Regression** on scatter plots and for processing **handwritten digits** and **character recognition** using the MNIST dataset
- **Maximized dispersion** in a 28x28 pixel image on an 84D Hyperplane through **Dimensionality Reduction**
- Performed **Reverse Image Processing** to get the original image from the 84-coordinate system and used the **Forbenious norm of difference** to measure closeness while processing and grouping similar images

OTHER PROJECTS

Rail Journey Planner | Course Project

(Autumn 2022)

Guide: Prof. Supratik Chakraborty, Course: Data Structures and Algorithms Lab

- Developed a **Railway Journey Planner** using **BST**, **queues**, **linked lists**, **dictionaries**, **trees**, and many more.
- Implemented algorithms like **Quicksort** for functionalities like sorting trains based on departure time, **KMP** for differentiating good and bad reviews, and **AVL** trees for balanced data storage
- Included features like **auto-complete** of station names and **filtered reviews** using **Tries** and **Max-Heaps**

Decentralized App to detect Fake News | Course Project

(Spring 2024)

Guide: Prof. Vinay Ribeiro, Course: Introduction to Blockchains, Cryptocurrencies and Smart Contracts

- Implemented DApp as a **smart contract** on a permissionless blockchain for **fact-checking** the news items
- Developed model to boost **voter trustworthiness** by incentivizing rational behavior and resisting **Sybil attacks**

Socket Programming | Course Project

(Spring 2023)

Guide: Prof. Bhaskaran Raman, Course: Computer Networks

- Implemented **Client-Server program** to transfer a file from the server to the client using **TCP sockets**
- On successful **bind and listen**, server establishes a connection with the client and undergoes transmission
- Enhanced the system to enable **Multiple connections** using the select system call by the **master socket**

Upgraded XV6 | Course project

(Autumn 2023)

Guide: Prof. Purushottam Kulkarni, Course: Operating Systems Lab

- Examined xv6 source code and integrated **process scheduling** and dynamical adjustment of **time slice quanta**
- Added functionality of **lazy page allocation** with **page fault handling**, reducing unnecessary memory usage
- Implemented **threading** and **robust synchronization** mechanisms to ensure concurrent execution of processes

POSITIONS OF RESPONSIBILITY

Design Convener | Institute Technical Council

(May '22 - May '23)

Selected to be a part of a **3** membered team out of **32** applicants to create elegant graphic content for **ITC** and provide assistance in the designing necessities.

- Coordinated with **10** council clubs and made Posters, T-shirts, Certificates, Banners, Wall paintings, and more
- Contributed to many design elements for major events such as **Inter IIT Tech Meet**, **Tech & RnD Expo**

TECHNICAL SKILLS

Programming	C++, Python, Go, Java, Bash, Flex, Bison, VHDL, MIPS, Sed, Awk
Data Science	MATLAB, NumPy, Pandas, Matplotlib, Scikit-learn
Software Development	Django, HTML, CSS, JavaScript, Bootstrap, PostgreSQL
Other Tools	Wireshark, GitHub, Z3solver, NodeJS, ExpressJS

KEY COURSES UNDERTAKEN

Computer Science	DSA, Algorithms, Computer Networks, Computer Architecture, Operating Systems, Compilers, Database and Information Systems, Blockchains and Cryptocurrencies, Automata Theory, Game Theory
Data Science	Data Analysis and Interpretation, Digital Image Processing, Artificial Intelligence and Machine Learning, Optimisation Models
Mathematics	Calculus, Linear Algebra, Differential Equations, Numerical Analysis, Probability and Stochastic Processes

EXTRACURRICULAR ACTIVITIES

- Completed **1 year** training in **NCC(National Cadet Corps)** from **2 Maharashtra Engineers Regiment**
- Took part in the **E-Summit'23** organized by the **Entrepreneurs Cell** of IIT-Bombay
- Participated in the **Versova Beach Cleanup** campaign organized by **Abhyuday**, IIT Bombay